INMO(23rd) – 2008

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Let ABC be a triangle, I is its incentre; A_1, B_1, C_1 be the reflections of I in BC, CA, AB respectively. Suppose the circum circle of triangle $A_1B_1C_1$ passes through A. Prove that B_1, C_1, I, I_1 are concyclic, where I_1 is the incentre of triangle $A_1B_1C_1$.
2. Find all triples (p, x, y) such that $p^x = y^4 + 4$, where p is a prime and x, y are natural numbers.
3. Let A be a set of real numbers such that A has at least four elements. Suppose A has a property that $a^2 + bc$ is a rational number for all distinct numbers a, b, c in A. Prove that there exists a positive integer M such that $a\sqrt{M}$ is a rational number for every a in A.
4. All the points with integer coordinates in the xy plane are coloured using three colours red, blue and green, each colour being used at least once. It is known that the point (0, 0) is coloured red and the point (0, 1) is coloured blue. Prove that there exist three points with integer coordinates of distinct colours which form the vertices of a right angled triangle.
5. Let ABC be a triangle $\Gamma_A, \Gamma_B, \Gamma_C$ be three equal disjoint circles inside ABC such that Γ_A touches AB, Γ_B touches BC and CA. Let Γ be a circle touching circles $\Gamma_A, \Gamma_B, \Gamma_C$ externally. Prove that the line joining the circumcentre O and the incentre I of triangle ABC passes through the centre of Γ.
6. Let P(x) be a given polynomial with integer coefficients. Prove that there exist two polynomials Q(x) and R(x), again with integer coefficients, such that (i) P(x)Q(x) is a polynomial in x^2 and (ii) P(x)R(x) is a polynomial in x^3.

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